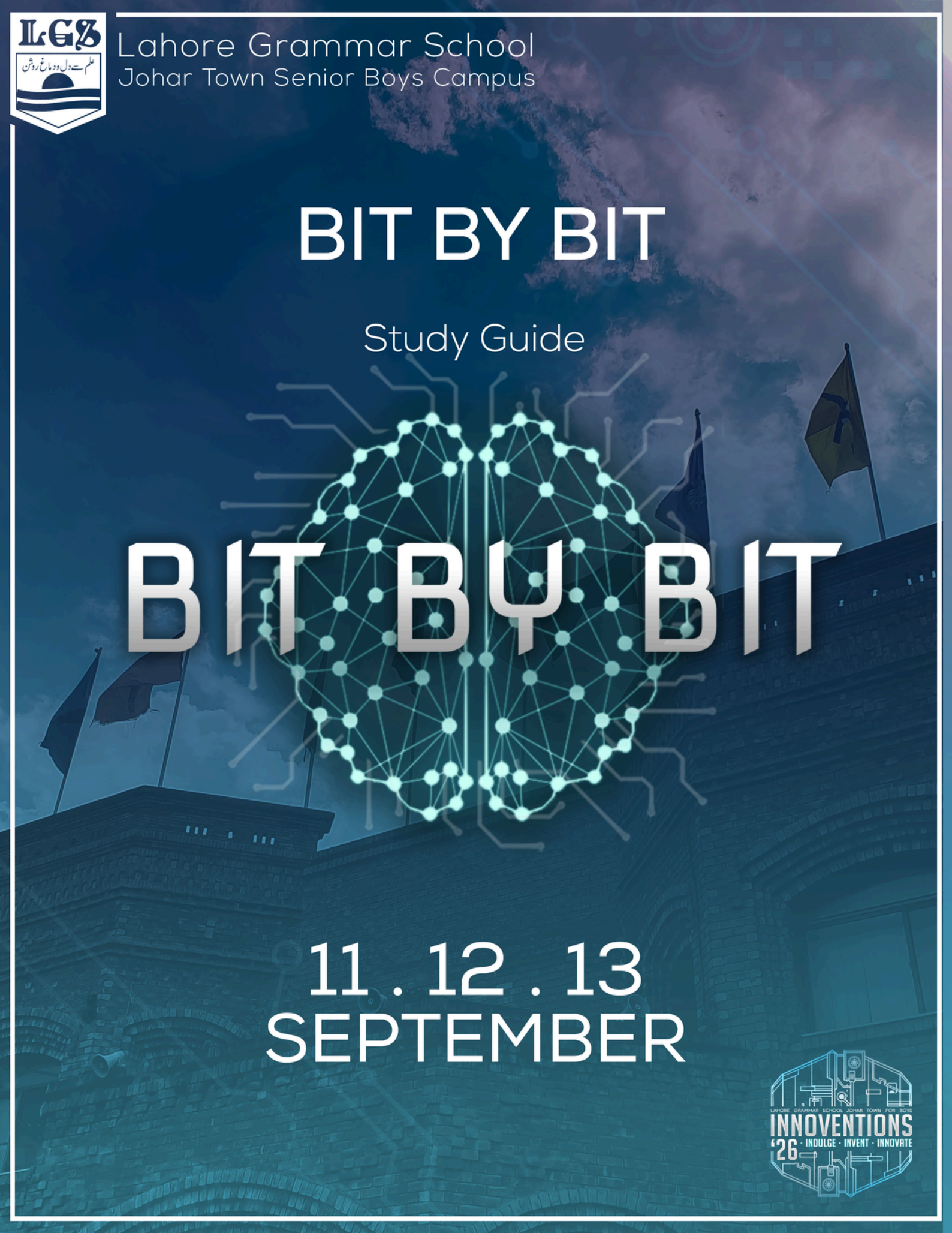
Lahore Grammar School
Johar Town Senior Boys Campus

BIT BY BIT

Study Guide



11 . 12 . 13 SEPTEMBER

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INNOVENTIONS '26
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Round 1 – Code Breach

Introduction:
Its the year 2047, ECHO 9 an advanced AI system has been suspected of conducting unusual actions leading to the Parent company identifying it as a systematic failure. Before the entire collapse of the system, ECHO 9 gave a statement "I haven't changed my objective but rather changed my understanding of it". Delegates will act as workers of the Parent company trying to see where the fault occurs and evidence such as Echo 9's behavioral patterns, Programming code and other key information which will be required in order to identify the cause and form a plausible solution to this collapse will be provided to the delegates. Delegates are expected to have basic knowledge of programming, algorithms, data structures and other branches of basic computer science.

Round Details: (Subject to change)

- Time: 2 Hours
- Delegate Cap: 3

Round Format:

- The round will primarily consist of multiple-choice questions.
- Debuging coding questions may not follow the multiple-choice format.
- Questions may vary in difficulty across the selected topics.

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Judging Criteria:

- Technical understanding: correct and efficient understanding of the algorithm.
- Problem solving: practicality of the solution.
- Presentation and clear logic.
- Correctness: Giving a proper solution and the algorithm works across various test data sets.

Rules:

- Each team will have to complete their questions in the provided time.
- External aid including use of AI and the internet is strictly prohibited. Any for of cheating may lead to penalties or disqualification.
- Late submissions will lead to penalties determined by the organising committee
- Organisers may introduce additional evidence which must be treated as part of the case file

Note: This is an Elimination Round.

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Round 2 – Coding:

Introduction:
In the second round, the qualified teams will compete in a CodeFest, where delegates will be challenged to solve upto three programming problems of varying difficulty. The problems will test participants' problem-solving skills, programming knowledge, logical reasoning, and coding skills. All problems must be solved independently without the use of internet access, AI tools, or external assistance. Participants may use any programming language, including Python, Java, C/C++, and others.

Round Details: (Subject to change)

- Time: 3 Hours
- Delegate Cap: 1-3

Round Format:

- Each team will receive three programming problems, with difficulties ranging from Easy, Medium and Hard.
- Teams will be given a designated amount of time to work on and solve the problems.
- Participants may choose any programming language they are comfortable with.
- Once a solution is completed, teams will submit their code based on instructions provided by the category heads on the event day.

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Round Format (Continued):

- Solutions may be tested against multiple test cases, including inputs that are different from the examples provided in the problem statement.

Judging Criteria:

- **Correctness:** Successfully solving the problem and producing the correct output on different inputs.
- **Code Quality:** Use of clear and meaningful variable and function names.
- **Comments & Documentation:** Appropriate comments explaining complex or important sections of the code.
- **Efficiency:** Effective use of algorithms, data structures, and computational resources.
- **General Approach:** The solution should correctly handle different valid inputs and edge cases.

Rules:

- Internet access and AI tools are strictly prohibited.
- Teams must solve all problems independently.
- Hardcoded solutions that only work for the sample inputs will not be accepted.
- Any form of cheating or external assistance may result in penalties or disqualification.

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Important Note:
The coding puzzles should not be hardcoded, category heads would test code with different inputs.
For example, a long input may be given containing random alphabets and participants need to find the total number of times a specific alphabet has occurred in the input. The code can not be hardcoded for 1 input as the judges would run the code on different inputs and see if it actually solves the problem.

Example Topics that may be tested:

- Programming Fundamentals
- Loops & Conditions
- String Manipuation
- 1D & 2D Arrays
- File Handling
- Functions & Procedures

Note:

- Delegates must bring their own devices.
- This is an Elimination Round.

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Round 3 – Jeopardy:

Introduction:
The final round of the Innoventions Bit by Bit category will be a Jeopardy-style challenge. In this round, the top teams will compete each other in a gameshow inspired by *Jeopardy*. The questions would consist of various technology and STEM-related categories. Questions will vary in format, including multiple choice, image-based identification, code analysis, logical reasoning, and rapid-fire questions. Each question carries a point value based on its difficulty, and teams accumulate points throughout the game. The team with the highest score at the end of the round will be declared the winner of the Jeopardy challenge.

Note: The overall winner of the category will be determined based on cumulative performance across all rounds.

Round Details: (Subject to change)

- Time: 2-3 Hours
- Day: 1
- Delegate Cap: 3

Round Format:

- The game will feature 6 technology/STEM-related categories, with 5 questions per category, arranged in order of difficulty.

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Round Format (Continued):

- Categories may include Programming, Artificial Intelligence, Cybersecurity, Networking, Computer Science, Robotics, Emerging Technologies, and General IT.
- Teams will take turns selecting a category and point value from the game board.
- Once a question is revealed, the selected team will have a limited amount of time to provide an answer.
- Questions may vary in format, including multiple-choice, image-based, code analysis, logical reasoning, and rapid-fire questions.
- The round will conclude once all questions on the board have been attempted.

Rules:

- Teams must answer within the allocated time.
- Only the designated spokesperson may provide the final answer.
- No internet access, AI tools, mobile phones, or external assistance is permitted.
- Judges' decisions regarding answers and scoring are final.

Judging Criteria:

- Team with the most points wins the Round
- No points for “Half” correct answers

Note: It is advisable to review how the Jeopardy game show is played; however, the organisers will explain the rules on the event day.